

# Amit Kumar Mohanty

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## PROFESSIONAL SUMMARY

Machine Learning Engineer with 1+ year of experience architecting production-grade AI systems, high-throughput model-serving platforms, and event-driven applications. Proven track record of leading technical initiatives, designing low-latency services at enterprise scale, and delivering measurable business impact through ML engineering and AI-driven automation. Outside of work, a competitive tennis player and avid sketch artist, pursuits that sharpen discipline, focus, and creativity.

## EDUCATION

### Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 – July 2028

M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing)

Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning

### K.R. Mangalam University

July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon

## EXPERIENCE

### Research Associate – AI, Keywords Studios India – Gurgaon, Haryana

May 2025 – Present

- Awarded “Rookie of the Year 2025” for surpassing all performance KPIs through advanced model training, data-pipeline optimization, and cross-functional collaboration.
- Trained and optimized enterprise-grade Agentic AI models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving task-completion reliability.
- Contributed to a cutting-edge web-automation AI agent that visually interprets on-screen content and, from a natural-language instruction, autonomously plans and executes browser actions such as clicking, typing, and navigating to complete complex, multi-step tasks.
- Trained, evaluated, and quality-controlled the agent with a global team, curating high-quality datasets and running structured test reviews to deliver client-ready iterations with improved action accuracy and reliability.
- Built custom data pipelines and delivered 85% of self-generated training datasets directly to production as the first member of a global team, accelerating model readiness.

### Artificial Intelligence Intern, Infosys – Remote

November 2024 – February 2025

- Engineered an AI call-center system with LangChain and LangSmith, boosting model response accuracy by 35% and elevating customer-AI interaction quality.
- Developed and deployed LLM features that reduced system latency by 25%, streamlining the model deployment process.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing hallucinations by 20% and making customer conversations feel more natural and trustworthy.
- Collaborated with a team of 12 engineers to integrate end-to-end ML workflows and earned selection to present at the Best Project by Indian Interns initiative.

## PROJECTS

### Dynamic Screen Companion

June 2025 – Present

- Built a real-time multi-modal assistant integrating the Gemini API to analyze live screen activity and deliver continuous voice feedback and interactive support.
- Engineered a high-performance FastAPI model-serving backend handling concurrent inference streams, reducing manual task time by 40%.
- Designed the end-to-end multi-modal architecture from scratch and optimized OCR pipelines to maintain 95%+ extraction accuracy across varied screen layouts.

### Diabetic Optiscan

January 2025 – June 2025

- Built an AI tool that scans retinal (eye) images to detect early signs of diabetic retinopathy, and co-authored a peer-reviewed research paper explaining how the system was designed and the accuracy improvements it achieved.
- Led integration of a Vision Transformer (ViT-B/16) model on the APTOS dataset, optimizing self-attention to capture complex spatial patterns in high-resolution retinal images.
- Designed a custom wavelet modification technique that boosted detection precision by 50% over legacy methods.

### Currency Detection App for Visually Impaired

July 2024 – December 2024

- Trained an image-classification model on 200,000+ banknote images achieving 96% accuracy, deployed in a mobile app with voice feedback to help prevent fraud.
- Conducted user interviews with visually impaired individuals to refine real-world accessibility features.

## SKILLS

**Languages & Frameworks:** Python, C++, SQL, R, FastAPI, TensorFlow, PyTorch, OpenCV, Scikit-learn, LangChain, LangGraph

**AI & Machine Learning:** LLMs, Generative AI, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Deep Learning, Transformers, Hugging Face, BERT, NLP, Prompt Engineering, Model Evaluation & Optimization, OOP, DSA

**MLOps & Deployment:** Docker, Kubernetes, CI/CD, MLflow, Model Serving, Model Monitoring, Data Pipelines, Git, Vector Databases (FAISS/Chroma)

**Data & Cloud Tools:** AWS, GCP (Vertex AI), Power BI, MySQL, MongoDB, Pandas, NumPy, Statistical Analysis

## CERTIFICATIONS

### Ducat India – Data Science Professional Training

April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

### AI/ML for Geodata Analysis – ISRO

Applied ML and deep learning to satellite and remote-sensing imagery for terrain classification and geospatial pattern analysis.